Network Automation Training(Upskill & Reskill)

Introduction:

Welcome to AimNxt, a leader in network training. Our mission is to provide exceptional, hands-on training in network automation and Python programming. This programme outlines a custom, instructor-led course designed to meet the evolving needs of today's network professionals.

The program is designed to provide individuals with the knowledge, skills, and hands-on networking and coding experience necessary to excel in the field of Network Automation.

Course Objective:

Our training program is meticulously crafted to empower network professionals. The core objective is to provide in-depth knowledge and hands-on experience in using tools like Ansible and Python, along with understanding common data formats. This course aims to skilfully guide attendees through the intricacies of automating network tasks, fostering both efficiency and innovation in their professional roles.

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Prerequisites

To ensure that all participants get the most out of our training, we recommend the following prerequisites:

- A basic understanding of network concepts, ideally with some practical experience.
- An introductory familiarity with the Linux operating system, including basic command-line usage.

Course Outline

Module 1: Introduction to Network Automation

- Real-World Use Cases for Network Automation
- Essential Components of an Automated Network
- Fundamental Principles and Best Practices

Module 2: Data Formats - JSON and YAML

- Understanding JSON: Structure and Components
- Exploring YAML: Structure and Components
- Techniques for Parsing JSON
- Techniques for Parsing YAML

Module 3: REST APIs

- Comprehensive Overview of REST APIs
- HTTP Essentials: Methods, Headers, and Responses
- Interpreting HTTP Status Codes
- Constructing REST Queries
- Authentication Strategies for REST Queries
- Effective Communication with REST APIs
- Practical Use of Postman for API Interactions
- Python Techniques for REST API Queries
- Utilization of NX-API for REST API Calls

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Module 4: Ansible for Network Automation

- Introduction to Ansible
- Managing Static Inventories in Ansible
- Creating and Executing Ansible Tasks
- Implementing Loops in Ansible
- Managing Variables in Ansible
- Organizing Host and Group Variables
- Utilizing Ansible Modules for Network Tasks
- Enhancing Automation with Lookup Plugins
- Applying Ansible Filters for Data Manipulation
- Combining Ansible with Jinja2 Templating
- Conducting Ansible Dry Run Testing
- Utilizing Check Mode for Pre-deployment Verification
- Gathering Network Facts with Ansible

Module 5: Introduction to Python and Netmiko

- Getting Started with Python for Networking
- Installation and Management of Python Packages
- Working with Virtual Environments
- Understanding Scripts, Modules, Packages, and Libraries
- Python Data Types and Their Applications
- Implementing Conditional Statements
- Mastering Looping and Iteration
- Handling User Inputs in Python Scripts
- Writing Reusable Functions
  Introduction to Netmiko for Network Device Interaction

Overview

AimNxt School of Tech Leaders

AimNxt Network Automation Enablement is the perfect start if you know you want to build, manage and support of enterprise IT infrastructure.

AimNxt Network Automation Course can helps you prepare for a wide variety of IT jobs, including:

- Software Engineer
- Network Automation Engineer
- Network programmer
- Network Developer
- Software Developer

Next Steps

AimNxt Network Automation enablement plays a crucial role in enabling developers to adapt to the changing landscape of networking and infrastructure technologies, making it easier for organizations to adopt automation, programmability, and software-defined networking to meet their business objectives. Whether you're a network engineer, software developer, or IT professional, Network Automation module provides the tools andresources to enhance your skills and build innovative solutions in the world of networking and IT infrastructure Demanding IT& Software career"